

Lecture 02

Vector II: Norm and Inner Product

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Recall

In the last lecture ...

- Definition of **(column) vectors**
- Basic vector operations (**addition, multiplication, linear combination**)

Linear Algebra Terminology and Conventions

- A **vector** is an **ordered list of numbers** (often a column) written as

$$\begin{bmatrix} -1.1 \\ 0.0 \\ 3.6 \\ -7.2 \end{bmatrix} \quad \text{or} \quad (-1.1, 0, 3.6, -7.2)$$

- Numbers in the list/array are the **elements** (**entries, coefficients, components**)
- Number of elements is the **size** (dimension) of the vector (n -dimensional)
e.g. The vector above has size 4; its third entry is 3.6
- Numbers are called **scalars** (compared to vectors)

Linear Algebra Terminology and Conventions

- Vectors are often represented by [mathematical symbols](#), e.g.,

$$x, y, u, v$$

It could be either in boldface or not, depending on contexts

Other conventions for beginners and engineers: **u**, **v** or $\overrightarrow{\mathbf{u}}$, $\overrightarrow{\mathbf{v}}$

- The [i-th element](#) of an n -vector **u** is often denoted as u_i

[[Warning](#)]: sometime u_i may be used to denote the i -th vector in a list of vectors

- Two vectors **u**, **v** are [equal](#) iff $\mathbf{u} - \mathbf{v} = \mathbf{0}$, written as $\mathbf{u} = \mathbf{v}$

Linear Algebra Terminology and Conventions

- **Zero vector**: All elements are zeros, e.g., $(0,0,0,0)$ denoted by $\mathbf{0}$, $\mathbf{0}$, or $\mathbf{0}_n$
- **One vector**: All elements are ones, e.g., $(1,1,1,1)$ denoted by $\mathbf{1}$, $\mathbf{1}$, or $\mathbf{1}_n$
- **Non-zero vector**: Not a zero vector

Learning Goals Today

Today: More vector operations (1) Vector Norm; (2) Inner Product.

After this lecture, you should be able to:

- **Logic:** If and only if
- **calculate the norm & inner product**
- **tell a few major properties of norm & inner product**
- **utilize the Cauchy Schwarz inequality and triangular inequality**
- **provide real-world applications of "inner product"**

Part 0 Preliminary: Logical Word “iff”

预备知识：逻辑学词汇“iff”

Basic Logic I: Implication

We will introduce a new word “iff”, used throughout the course.

Let's start from simple logic.

Remark: This part is not contained in textbooks.

- **Statement:** "If it rains (P), then the ground is wet (Q)."
- **Notation:** $P \Rightarrow Q$
- **Key Concept:** This is a **one-way street**.
- **Meaning:** Rain is *sufficient* to make the ground wet.

The arrow points one way.

What happens when we try to go backward?

Basic Logic II: Reverse

Question: If the ground is wet (Q), did it necessarily rain (P)?

Answer: No.

(It could be car wash, a spilled bucket, etc.)

Just because the ground is wet doesn't mean it rained.

- **Notation:** $Q \not\Rightarrow P$
- **Takeaway:** The reverse direction is **not** automatically true.

Basic Logic III: Two Way Streets

Sometimes, the logic works perfectly in both directions.

- **Example:**

- P : "Today is Saturday."
- Q : "Tomorrow is Sunday."

- **Logic:**

1. If today is Saturday $\Rightarrow$ Tomorrow is Sunday. (TRUE)
2. If tomorrow is Sunday $\Rightarrow$ Today is Saturday. (TRUE)

Notation: $P \Leftrightarrow Q$.

It reads "P is equivalent to Q".

Starting from today, we also say "P **iff** Q".

It reads "P **if and only if** Q"

Next, we explain "only if".

Basic Logic IV: If & Only If

If it rains (P), then the ground is wet (Q)

If P, then Q

$P \Rightarrow Q$

Equivalent statement:

The ground is wet (Q) if it rains (P).

Q If P

$P \Rightarrow Q$

Equivalent statement:

It rains (P) only if the ground is wet (Q).

P only if Q.

$P \Rightarrow Q$

English Phrase	Arrow Direction	Logic
If A, then B	Forward	$A \Rightarrow B$
A only if B	Forward	$A \Rightarrow B$
A if B	Backward	$B \Rightarrow A$

Easy to make mistakes!

Judgement Exercise

Judge: True or False.

(1) "P only if Q" just means " $Q \Rightarrow P$ " $P \Rightarrow Q$. False

(2) The ground is wet only if it rains. False.

(3) Today is Monday only if tomorrow is Tuesday. True.

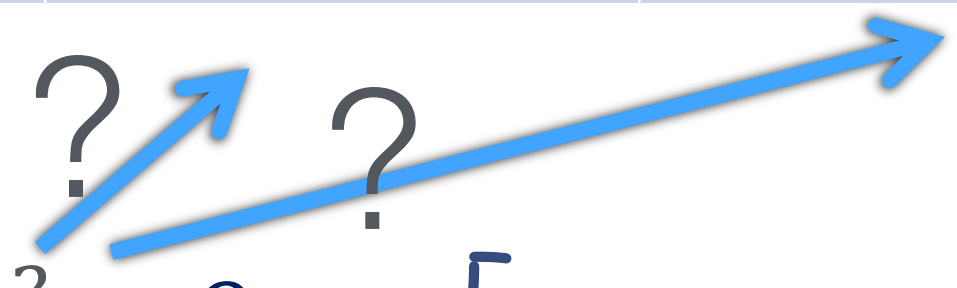
(4) $x = 3$ only if $x^2 = 9$. True.

i.e. $x=3 \Rightarrow x^2=9$ ✓

Basic Logic V: Math Examples of If and Only If

Math Notation	Statement in Language	Example of true statement	Example of false statement
$P \Rightarrow Q$	If P, then Q or Q if P	$x^2 = 9$ if $x = 3$,	$x^2 = 6$ if $x = 3$
$Q \Rightarrow P$	Q only if P or equivalently, if Q, then P	$x = 3$ only if $x^2 = 9$ Or equivalently, if $x = 3$, then $x^2 = 9$.	$x^2 = 9$ only if $x = 3$
$P \Leftrightarrow Q$	P iff Q.	②	①

Exercise

- ① $x = 3$ iff $x^2 = 9$? F
- ② $x = 3$ iff $2x = 6$? T
- 

Basic Logic VI: Proving “Iff”

To prove P iff Q , you need to prove:

First, $Q \Rightarrow P$; or the “if” part.

Second, $P \Rightarrow Q$; or the “only if” part.

Exercise: Prove: $x = 1$ iff $2x^2 - 4x + 2 = 0$.

Proof:

“If” Part:

Need to prove $x=1$ if $2x^2-4x+2=0$

$$2x^2 - 4x + 2 = 0 \Rightarrow 2(x-1)^2 = 0 \Rightarrow x-1=0 \Rightarrow x=1.$$

“Only if” part: *Need to prove $x=1$ only if $2x^2-4x+2=0$.*

$$x=1 \Rightarrow 2x^2 - 4x + 2 = 2 \cdot 1^2 - 4 \cdot 1 + 2 = 0.$$

Make sure you can solve the exercise.
Otherwise you may be lost later.

Remark: Sometimes you can directly prove by a bunch of “ $\Leftrightarrow$ ”.

$$2x^2 - 4x + 2 = 0 \Leftrightarrow 2(x-1)^2 = 0 \Leftrightarrow x-1=0 \Leftrightarrow x=1.$$

But sometimes you have to prove two directions separately

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Common issue 1.

(())

if part: $x = 1 \Rightarrow 2x^2 - 4x + 2 = 2 \cdot 1^2 - 4 \cdot 1 + 2 = 0$

One way to get around the issue

restate: Prove $x = 1 \Leftrightarrow 2x^2 - 4x + 2 = 0$.

① " $\Rightarrow$ " part: $x = 1 \Rightarrow \dots$

② " $\Leftarrow$ " part: If $2x^2 - 4x + 2 = 0$, then blabla...

However sometimes restating the desired result is too long.

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Common issue 2

"If part": $2x^2 - 4x + 2 > 0$

$\Rightarrow$ $x^2 - 2x + 1 = 0$
 $\Rightarrow (x-1)^2 = 0$
 $x = 1$

missing logical relation words

Words/symbols for logical relation

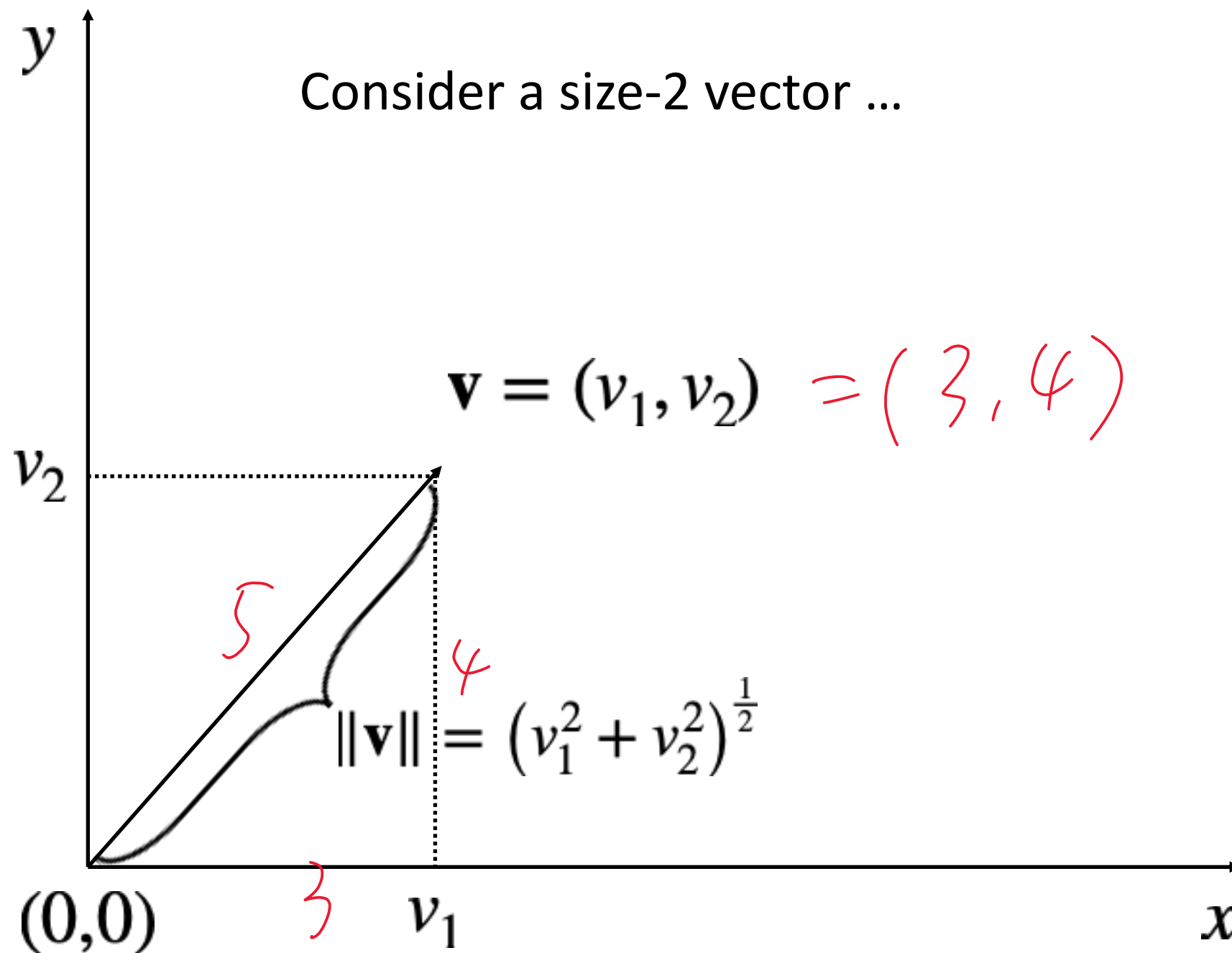
① $\Rightarrow$

② If ..., then ...
thus ... so ...

③ $\therefore$

Part I Vector Norm

Vector Norm (“Geometric Length”)



Vector Norm

A generalized notion of “*absolute value*” ...

Definition (norm)

Let $\mathbf{v} = (v_1, \dots, v_n)$ be a vector. The **norm** of $\mathbf{v}$, denoted by $\|\mathbf{v}\|_2$ is defined as

$$\|\mathbf{v}\|_2 := (v_1^2 + \dots + v_n^2)^{\frac{1}{2}}$$

The norm is also called an *Euclidean norm* or geometric length.

We often abbreviate $\|\cdot\|_2$ as $\|\cdot\|$

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Properties of Vector Norms

Suppose $\mathbf{v}$ is a vector.

Property 1 (P1) $\|\mathbf{v}\| \geq 0$, and $\|\mathbf{v}\| = 0$ iff $\mathbf{v} = \mathbf{0}$.

Property 2 (P2) If c is a scalar, then $\|c\mathbf{v}\| =$

Interpretation

Examples of P1:

$\|\mathbf{v}\| \geq 0$ arbitary $\vec{v} = \begin{bmatrix} 1 \\ -4 \\ -7 \end{bmatrix}$, we know $\|\vec{v}\| \geq 0$, without computing

$\|\mathbf{v}\| = 0$ if $\mathbf{v} = \mathbf{0}$: $\vec{v} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$, we know $\|\vec{v}\| = 0$

$\|\mathbf{v}\| = 0$ only if $\mathbf{v} = \mathbf{0}$: If $\|\vec{v}\| = 0$, we know for sure $\vec{v} = \mathbf{0}$.

Examples of P2:

$c = 8$, $\vec{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$, $c\vec{v} = \begin{bmatrix} 24 \\ 32 \end{bmatrix}$, $\|c\vec{v}\| = \sqrt{24^2 + 32^2}$
too heavy

easier way: $\|c\vec{v}\| = 8 \cdot \sqrt{3^2 + 4^2} = 8 \cdot 5 = 40$.

Proof of Properties of Vector Norms

Class Break

Suppose $\mathbf{v}$ is a vector.

Property 1 ^① $\|\mathbf{v}\| \geq 0$, and ^② $\|\mathbf{v}\| = 0$ iff $\mathbf{v} = \mathbf{0}$.

Proof of P1:

① $\|\vec{v}\| = \sqrt{v_1^2 + \dots + v_n^2} \geq 0$ since square root ≥ 0 by definition.

② Need to prove $\|\vec{v}\| = 0$ iff $\vec{v} = \mathbf{0}$.

"If" part: $\vec{v} = \mathbf{0} \Rightarrow \|\vec{v}\| = \sqrt{v_1^2 + \dots + v_n^2} = \sqrt{0^2 + \dots + 0^2} = 0$.

"Only if" part. $\|\vec{v}\| = 0 \Rightarrow \sqrt{v_1^2 + \dots + v_n^2} = 0$

$$\Rightarrow v_1^2 + \dots + v_n^2 = 0$$

$$\stackrel{(*)}{\Rightarrow} v_1 = \dots = v_n = 0. \quad \square$$

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More complete proof.

(*) part: $v_1^2 + \dots + v_n^2 = 0$. ①

We know $v_1^2 \geq 0, \dots, v_n^2 \geq 0$.

Assume for the contrary that some $v_i^2 > 0$,
then $v_1^2 + \dots + v_n^2 \geq v_i^2 > 0$, contradicting ①.

Thus $v_1^2 = \dots = v_n^2 = 0$. ②

Shorten: ① $\Rightarrow$ ②,

Proof of Properties of Vector Norms

Suppose v is a vector.

Property 2 (P2) If c is a scalar, then $\|cv\| = |c| \|v\|$

Proof of P2: *def of $c \cdot \vec{v}$ & $\|\cdot\|$*

$$\|(-3) \cdot v\| = 3 \cdot \|v\|$$

$$\|c \vec{v}\| = \sqrt{(cv_1)^2 + \dots + (cv_n)^2}$$

associative rule

$$= \sqrt{c^2 (v_1^2 + \dots + v_n^2)}$$

$$= \sqrt{c^2} \cdot \sqrt{v_1^2 + \dots + v_n^2}$$

def.

$$= |c| \cdot \|v\|$$

Unit Vector (单位向量)

Definition (Unit Vector)

A vector $\mathbf{v}$ is called a **unit vector** if $\|\mathbf{v}\| = 1$

Examples:

$$\vec{v} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\vec{v} = \frac{1}{5} \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} \frac{3}{5} \\ \frac{4}{5} \end{bmatrix},$$

$$\|\vec{v}\| = \sqrt{\frac{3^2 + 4^2}{5^2}} = 1,$$

$$\vec{v} = \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \\ 0 \end{bmatrix}.$$

A Claim on Unit Vector (单位向量)

Claim: For any non-zero vector $\mathbf{v}$, $\frac{\mathbf{v}}{\|\mathbf{v}\|}$ is a unit vector.

Exercise: Prove it.

Proof Method 1 (using P1)

$$\text{Let } c = \frac{1}{\|\vec{v}\|} \text{ ① } \quad \vec{u} = \frac{\vec{v}}{\|\vec{v}\|} = c \cdot \vec{v},$$

$$\|\vec{u}\| \stackrel{\text{P1}}{=} \|c \cdot \vec{v}\| \stackrel{\text{①}}{=} \frac{1}{\|\vec{v}\|} \cdot \|\vec{v}\| = 1.$$

Proof Method 2 (by definitions)

$$\frac{\vec{v}}{\|\vec{v}\|} = \left(\frac{v_1}{\|\vec{v}\|}, \dots, \frac{v_n}{\|\vec{v}\|} \right),$$

$$\Rightarrow \left\| \frac{\vec{v}}{\|\vec{v}\|} \right\| = \sqrt{\left(\frac{v_1}{\|\vec{v}\|} \right)^2 + \dots + \left(\frac{v_n}{\|\vec{v}\|} \right)^2} = \sqrt{\frac{v_1^2 + \dots + v_n^2}{\|\vec{v}\|^2}} = 1.$$

Unit Vector: Examples

Examples (Unit Vector)

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

Standard Unit Vectors in a Cartesian coordinate system

[*Warning*]: In some textbooks, like Stephen Boyd, *only above vectors* are called unit vectors

Part II Inner Product

Vector Operations

Question 1: Can you think about any other vector operations?

Vector Addition

$$\begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} + \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix} = \begin{bmatrix} v_1 + w_1 \\ v_2 + w_2 \\ \vdots \\ v_n + w_n \end{bmatrix}$$

How to define product of vectors?

$\vec{v} \times \vec{u}$?

Vector Operations

Question 1: Can you think about any other vector operations?

Vector Addition

$$\begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} + \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix} = \begin{bmatrix} v_1 + w_1 \\ v_2 + w_2 \\ \vdots \\ v_n + w_n \end{bmatrix}$$

How about

$$\begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} \times \begin{bmatrix} w_1 \\ w_2 \\ \vdots \\ w_n \end{bmatrix} = \begin{bmatrix} v_1 w_1 \\ v_2 w_2 \\ \vdots \\ v_n w_n \end{bmatrix} \quad ?$$

This is a “point-wise product”.

But in this class, we often use another product.

Dot Product or inner product

Definition (Dot Product)

A *dot product* between two vectors (of the same size)

$$\mathbf{v} = (v_1, \dots, v_n)$$

$$\mathbf{w} = (w_1, \dots, w_n)$$

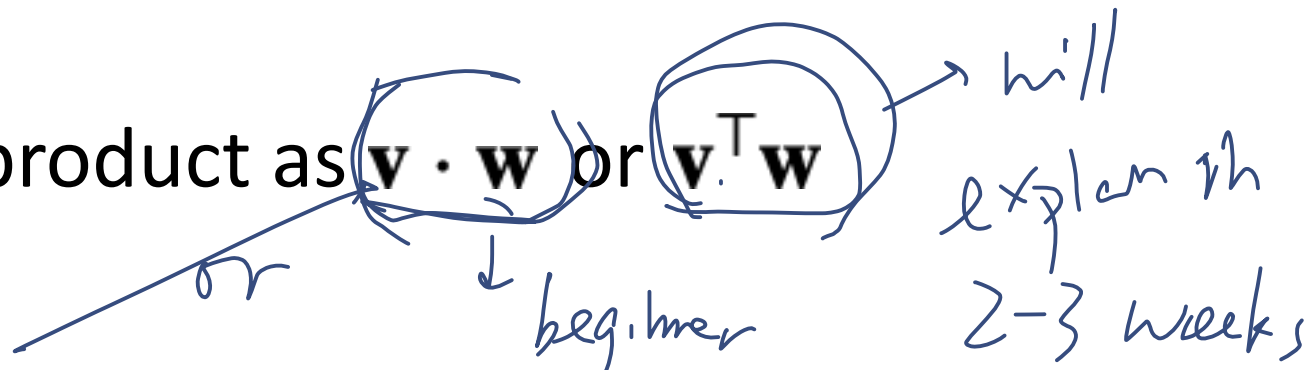
is defined as

$$\langle \mathbf{v}, \mathbf{w} \rangle = \sum_{i=1}^n v_i w_i$$

Example: Calculate the dot product of $(-1, 2, 2)$ and $(1, 0, -3)$

$$(-1) \times 1 + 2 \times 0 + 2 \times (-3) = -1 + 0 - 6 = -5.$$

Dot Product or Inner Product

Sometime we also write a dot product as $\mathbf{v} \cdot \mathbf{w}$ or $\mathbf{v}^T \mathbf{w}$ 
Recommend writing as $\langle \mathbf{v}, \mathbf{w} \rangle$ to avoid confusion.

A *dot product* is called an *inner product* in more general settings

Properties of Inner Products

Properties of Inner Products

P1 Linearity $\langle a\mathbf{v} + b\mathbf{u}, \mathbf{w} \rangle = a\langle \mathbf{v}, \mathbf{w} \rangle + b\langle \mathbf{u}, \mathbf{w} \rangle$ for all $\mathbf{u}, \mathbf{v}, \mathbf{w}$

$\langle \mathbf{v} + \mathbf{u}, \mathbf{w} \rangle = \langle \mathbf{v}, \mathbf{w} \rangle + \langle \mathbf{u}, \mathbf{w} \rangle$. $\langle \text{LC of } \vec{v}, \vec{u}, \vec{w} \rangle$ LC of dot product

Properties of Inner Products

Properties of Inner Products

P1 **Linearity** $\langle a\mathbf{v} + b\mathbf{u}, \mathbf{w} \rangle = a\langle \mathbf{v}, \mathbf{w} \rangle + b\langle \mathbf{u}, \mathbf{w} \rangle$ for all $\mathbf{u}, \mathbf{v}, \mathbf{w}$

P2 **Symmetry** $\langle \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{w}, \mathbf{v} \rangle$ for all $\mathbf{v}, \mathbf{w}$

对称性

Properties of Inner Products

Properties of Inner Products

P1 **Linearity** $\langle a\mathbf{v} + b\mathbf{u}, \mathbf{w} \rangle = a\langle \mathbf{v}, \mathbf{w} \rangle + b\langle \mathbf{u}, \mathbf{w} \rangle$ for all $\mathbf{u}, \mathbf{v}, \mathbf{w}$

P2 **Symmetry** $\langle \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{w}, \mathbf{v} \rangle$ for all $\mathbf{v}, \mathbf{w}$

P3 **Positivity** $\langle \mathbf{v}, \mathbf{v} \rangle \geq 0$ for all $\mathbf{v}$

$\langle \mathbf{v}, \mathbf{v} \rangle = 0$ iff $\mathbf{v} = \mathbf{0}$

Properties of Inner Products

Properties of Inner Products

P1 **Linearity** $\langle a\mathbf{v} + b\mathbf{u}, \mathbf{w} \rangle = a\langle \mathbf{v}, \mathbf{w} \rangle + b\langle \mathbf{u}, \mathbf{w} \rangle$ for all $\mathbf{u}, \mathbf{v}, \mathbf{w}$

P2 **Symmetry** $\langle \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{w}, \mathbf{v} \rangle$ for all $\mathbf{v}, \mathbf{w}$

P3 **Positivity** $\langle \mathbf{v}, \mathbf{v} \rangle \geq 0$ for all $\mathbf{v}$ $\langle \vec{v}, \vec{v} \rangle = v_1^2 + \dots + v_n^2$

$$\langle \mathbf{v}, \mathbf{v} \rangle = 0 \quad \text{iff} \quad \mathbf{v} = \mathbf{0}$$

verify $\langle \mathbf{v}, \mathbf{v} \rangle = \|\mathbf{v}\|^2$! $\langle \vec{v}, \vec{v} \rangle = v_1^2 + \dots + v_n^2 = \|\vec{v}\|^2$

Reading: Inner product naturally induces a norm and every “inner product space” is a “normed vector space”

Application 1: Weight, Feature and Score

Example 1.1: My MAT2041 final score

	Assignment & Quiz & Attendance	Midterm	Final	Total
Weight	w_1 0.35	w_2 0.3	w_3 0.35	
Score	v_1 90	v_2 85	v_3 95	
Cost	$w_1 v_1$ 31.5	$w_2 v_2$ 25.5	$w_3 v_3$ 33.25	90.25

$w = (0.35, 0.3, 0.35)$: weight vector

$v = (90, 85, 95)$: feature vector

$\langle v, w \rangle = 90.25 =: \text{score}$

$w_1 v_1 + \dots + w_3 v_3 = \langle \vec{v}, \vec{w} \rangle$

Controlled by instructor

Controlled by students

“weight vector”

“Feature vector”

“Score”

Application 1: Weight, Feature and Score

Example 1.2: Movie preference

	Action Film	Hollywood	Comedy	Total		
“weight”	Weight	10%	10%	80%	Preference of Adam	
“Feature”	Value	0.5	0.5	10	Movie 1: Moon Man	
	Score	0.05	0.05	8	8.1	“Score”



Application 1: Weight, Feature and Score

Example 1.2: Movie preference



Movie 2: The Matrix

“weight”

“Feature”

	Action Film	Hollywood	Comedy	Total
Weight	10%	10%	80%	
Value	10	8	2	
Score	7	1.6	0.2	8.8

Preference of Adam

Movie 2: Matrix

“Score”

Application 1: Weight, Feature and Score

Application 1: (general)

$\mathbf{w}$ is a vector of the same size (often called a **weight** vector),

$\mathbf{v}$ represents a set of “**features**” of an object,

Score: inner product $\langle \mathbf{v}, \mathbf{w} \rangle$ is a **weighted sum** of the feature values.

Examples:

	Grading	Movie	Your example?
“Weight”	Grading Scheme	Personal Preference	?
“Feature”	Scores	Movie Features	?
“Score”	Total Score	Score	?

Application 1: Weight, Feature and Score

Meta-application: evaluation.
(Useful for ranking, comparing, etc.)

Goal: To evaluate a city, a university, an employee, a basketball player, etc.

Step 1: Set up “rule”; specific to the evaluator

Step 1.1 Set up “features” (indicators)

Step 1.2 Provide **weights** for different features.

Step 2: Scoring each feature, for each object.

Provide a “rating” for each feature, obtaining a feature vector

Step 3: Compute “evaluation score”

by computing the inner product of weight vector and feature vector

Remark: “Score” can be used in other areas, e.g. machine learning

Summary Today

Today, we have learned:

--"If and only if". P only if Q means $P \Rightarrow Q$. P iff Q means $P \Leftrightarrow Q$

— Norm of vector: $\|\mathbf{v}\| = \|\mathbf{v}\|_2 := (v_1^2 + \dots + v_n^2)^{\frac{1}{2}}$

Properties of norm:

P1 $\|\mathbf{v}\| \geq 0$, and $\|\mathbf{v}\| = 0$ iff $\mathbf{v} = \mathbf{0}$.

P2 If c is a scalar, then $\|c\vec{\mathbf{v}}\| = |c| \|\vec{\mathbf{v}}\|$

— Inner product of two vectors

Properties: linearity, symmetry, positivity.

$$\langle \mathbf{v}, \mathbf{w} \rangle = \sum_{i=1}^n v_i w_i$$

— Applications of inner product:

“feature” and “score” for evaluation